

Solution No 1:

Maximize

$$Z = 3x_1 + 5x_2 + 4x_3 + 0.S_1 + 0.S_2 + 0.S_3$$

Subject to

$$2x_1 + 3x_2 + 0.x_3 + S_1 = 8$$

$$0.x_1 + 2x_2 + 5x_3 + S_2 = 10$$

$$3x_1 + 2x_2 + 4x_3 + S_3 = 15$$

$$x_1, x_2, x_3, S_1, S_2, S_3 \geq 0$$

Tabular Form:

Basic Variable	Z	x ₁	x ₂	x ₃ ,	S ₁	S ₂	S ₃	RHS
-	1	-3	-5	-4	0	0	0	0
S ₁	0	2	3	0	1	0	0	8
S ₂	0	0	2	5	0	1	0	10
S ₃	0	3	2	4	0	0	1	15

Solution No 2:

$$x_1 + x_2 + S_1 = 50$$

$$x_1 + 2x_2 - S_2 + A_1 = 80$$

$$3x_1 + 2x_2 - S_3 + A_2 = 140$$

$$x_1, x_2, S_1, S_2, S_3, A_1, A_2 \geq 0$$

Putting decision variable and slack equals to zero

$$x_1 = x_2 = S_2 = S_3 = 0$$

$$S_1 = 50$$

$$A_1 = 80$$

$$A_2 = 140$$

Initial Basic solution is feasible.

Solution No 3:

$$\text{Maximize } Z = -x_1 + 3x_2 - 2x_3$$

Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

Solution No 4:

Basic Variable	Z	x_1	x_2	x_3	S_1	S_2	S_3	RHS
-	1	1/3	0	-4	5/3	0	0	40/3
x_2	0	2/3	1	0	1/3	0	0	8/3
S_2	0	-4/3	0	5	-2/3	1	0	14/3
S_3	0	5/3	0	4	-2/3	0	1	29/3

Solution No 5:

$$\text{Minimize } Z = 3x + 2y + 7z + MA_1 + MA_2$$

subject to

$$-x + y + A_1 = 10$$

$$2x - y + z - S_1 + A_2 = 10$$

$$x, y, z, S_1, A_1, A_2 = 10$$

$$A_1 = 10 + x - y$$

$$A_2 = 10 - 2x + y - z + S_1$$

$$Z = 3x + 2y + 7z + M(10 + x - y) + M(10 - 2x + y - z + S_1)$$

After simplification we get

$$Z = (3 - M)x + 2y + z(7 - M) + MS_1 + 20M$$

$$\text{Setting } x = y = S_1 = 0$$

$$A_1 = 10$$

$$A_2 = 10$$

$$Z_{\min} = 20M$$

Solution No 2:

$$\text{Minimize } Z = A_1 + A_2$$

Subject to

$$2x_1 + x_2 + A_1 = 4$$

$$x_1 + 7x_2 + A_2 = 7$$

$$x_1, x_2, A_1, A_2 \geq 0$$

Solution No 3:

$$\text{Minimize } Z = A_1 + A_2$$

Subject to

$$3x + y + A_1 = 3$$

$$4x + 3y - S_1 + A_2 = 6$$

$$x + 2y + S_2 = 4$$

$$x, y, A_1, A_2, S_1, S_2 \geq 0$$

where

$$A_1 = 3 - 3x - y$$

$$A_2 = 6 - 4x - 3y + S_1$$

Since the objective function is expressed in terms of basic variables so to transform these into non basic Variables A_1 and A_2 from constraints into objective function.

$$Z = 3 - 3x - y + 6 - 4x - 3y + S_1$$

$$Z = 9 - 7x - 4y + S_1$$

Solution No 4:

$$\text{Minimize } Z = A_1 + A_2$$

subject to

$$2x_1 + x_2 + 2x_3 + A_1 = 4$$

$$3x_1 + 3x_2 + x_3 + A_2 = 3$$

$$x_1, x_2, x_3, A_1, A_2 \geq 0$$

Solution No 1:**Maximization:**

- 1) Objective function is of maximization type.
- 2) Inequalities are of $\leq$ type.
- 3) Slack variables are added to convert inequality into an equation.

Minimization:

- 1) Objective function is of minimization type.
- 2) Inequalities are of $\geq$ type.
- 3) Slack variables are subtracted and artificial variables are added to convert inequalities into an equation.

Solution No 2:

Maximize $Z = x_1 + x_2 + x_3$

subject to

$$x_1 + x_2 + S_1 = 8$$

$$-x_2 + x_3 + S_2 = 0$$

$$x_1, x_2, x_3, S_1, S_2 \geq 0$$

Setting $x_1 = x_2 = x_3 = 0$

$$S_1 = 8$$

$$S_2 = 0$$

One of the basic variable is zero hence the given solution is degenerate.

Solution No 3:

Transformed objective function is

$$\text{Maximize } Z = 2x_1 - 3x_2$$

subject to

$$x_1 - 3x_2 \leq 1$$

$$2x_1 - x_2 = 0$$

$$x_1, x_2 \geq 0$$

Including artificial variable and slack variable

$$x_1 - 3x_2 + S_1 = 1$$

$$2x_1 - x_2 + A_1 = 0 \Rightarrow A_1 = y - 2x$$

$$x_1, x_2, S_1, A_1 \geq 0$$

$$\text{Maximize } Z = 2x_1 - 3x_2 - MA$$

$$= 2x_1 - 3x_2 - M(x_2 - 2x_1)$$

$$= 2x_1 - 3x_2 - Mx_2 + 2Mx_1$$

$$= 2(1 + M)x_1 - (3 + M)x_2$$

$$\text{Setting } x_1 = x_2 = 0$$

$$S_1 = 1$$

$$A_1 = 0$$

One of the basic variable is zero hence the given solution is degenerate.

Solution No 1:

Maximize $Z = 2x_1 + 3x_2 + 10x_3$

subject to

$$x_1 + 2x_3 + A_1 = 0$$

$$x_2 + x_3 + A_2 = 1$$

$$x_1, x_2, x_3, A_1, A_2 \geq 0$$

Setting $x_1 = x_2 = x_3 = 0$

$$A_1 = 0$$

$$A_2 = 1$$

One of the basic variable is zero in the initial stage, so it is degenerate basic solution.

Solution No 3:

B.variable	x_1	x_2	S_1	S_2	S_3	Solution
Z	0	0	5/8	1/8	0	17/2
x_2	0	1	1/2	-1/2	0	2
x_1	1	0	-1/8	3/8	0	3/2
S_3	0	0	1	-2	1	4

Solution is temporarily degenerate

Solution No 4:

Tie between S_2 and S_3

B.variable	Z	x_1	x_2	S_1	S_2	S_3	RHS
	1	-2	0	0	5	0	150
S_1	0	1	0	1	0	0	40
x_2	0	0	1	0	1	0	30
S_3	0	1	0	0	-1	1	0

Second iteration:

B.variable	Z	x_1	x_2	S_1	S_2	S_3	RHS
	1	0	0	0	3	2	150
S_1	0	0	0	1	1	-1	40
x_2	0	0	1	0	1	0	30
x_1	0	1	0	0	-1	1	0

Basic variable x is equal to zero. This implies degenerate optimal solution.

Solution No 3:

$$x_1 = x_1' - x_1''$$

$$x_2 = x_2' - x_2''$$

$$Z = 2(x_1' - x_1'') + 3(x_2' - x_2'') + 0.S_1 + 0.S_2 + 0.S_3$$

s.t

$$-x_1' + x_1'' + 2x_2' - 2x_2'' + S_1 = 4$$

$$x_1' - x_1'' + x_2' - x_2'' + S_2 = 6$$

$$x_1' - x_1'' + 3x_2' - 3x_2'' + S_3 = 9$$

Taking $x_1' = x_1'' = x_2' = x_2'' = 0$

$$S_1 = 4$$

$$S_2 = 6$$

$$S_3 = 9$$

Solution No 4:

B.variable	Z	x ₁	x ₂	S ₁	S ₂	A	RHS
	1	-3-M	-4-2M	0	M	0	-12M
S ₁	0	2	1	1	0	0	4
A	0	1	2	0	-1	1	12

Second iteration

B.variable	Z	x ₁	x ₂	S ₁	S ₂	A	RHS
	1	M+1	0	4+2M	M	0	- 4M+16
x ₂	0	2	1	1	0	0	4
A	0	-3	0	-2	-1	1	4

We cannot proceed further with maximization as there is no negative coefficient in the objective row but the final solution has the artificial variable in the basis with the value equal to 4. Hence the problem has no feasible optimal solution.

Solution No 5:

Basic Variable	X₁	X₂	X₃	X₄	X₅	R	Solution
Z	0	-10	19	0	20	0	200
X ₄	0	-3	2	1	-3	0	20
X ₁	1	0	1	0	1	0	10
X ₆	0	-1	3	0	-1	1	10

Because the elements of pivot column are zero and negative, so the ratio does not have a proper meaning. So the solution is unbounded.

Solution No 1:

First we will write the standard form of the primal .Since the problem is of maximization all the constraints must involve the sign of $\leq$.

$$2w_1 + w_2 - w_3 \leq 2$$

Multilpying the second constraint by "-1"

$$2w_1 - w_2 + 5w_3 \leq 4$$

$$4w_1 + w_2 + w_3 \leq 6$$

$$w_1, w_2, w_3 \geq 0$$

Dual :

$$\text{Minimize } Z' = 2x_1 + 4x_2 + 6x_3$$

Subject to

$$2x_1 + 2x_2 + 4x_3 \geq 1$$

$$x_1 - x_2 + x_3 \geq 2$$

$$-x_1 + 5x_2 + x_3 \geq 1$$

$$x_1, x_2, x_3 \geq 0$$

Solution No 2:

In matrix form it v=can be written as

$$\text{Minimize } Z = (3, 1)[x_1, x_2] = cx$$

Subject to

$$\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A \quad x \quad \geq \quad b$$

Therefore the dual of the given problem is

$$\text{Maximize } Z' = b'w$$

$$= (2 \quad 1)[w_1 \quad w_2]$$

$$= 2w_1 + w_2$$

$$\text{subject to} \quad A'w \leq c'$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

or

$$2w_1 + w_2 \leq 3$$

$$3w_1 + w_2 \leq 1$$

$$w_1, w_2 \geq 0$$

Solution No 3:

Since it is of minimization problem all the constraints must involve the sign of $\geq$.

$$\text{Minimize } Z = 2x_2 + 5x_3$$

subject to

$$x_1 + x_2 \geq 2$$

multiplying the second constraint by -1.

$$-2x_1 - x_2 - 6x_3 \geq -6$$

Second constraint equation is equivalent to

$$x_1 - x_2 + 3x_3 \geq 4$$

$$-x_1 + x_2 - 3x_3 \geq -4$$

Hence the standard form of the primal is

$$\text{Minimize } Z = 2x_2 + 5x_3$$

$$x_1 + x_2 \geq 2$$

$$-2x_1 - x_2 - 6x_3 \geq -6$$

$$x_1 - x_2 + 3x_3 \geq 4$$

$$-x_1 + x_2 - 3x_3 \geq -4$$

Dual:

Dual of the given problem is

$$\text{Minimize } Z = 2x_2 + 5x_3 = (0 \quad 2 \quad 5) \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = cx$$

$$\begin{bmatrix} 1 & 1 & 0 \\ -2 & -1 & -6 \\ 1 & -1 & 3 \\ -1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 2 \\ -6 \\ 4 \\ -4 \end{bmatrix}$$

$$Ax \geq b$$

$$\text{Maximize } Z' = b'w = (2 \quad -6 \quad 4 \quad -4) \begin{bmatrix} w_1 & w_2 & w_3' & w_4' \end{bmatrix}$$

$$= 2w_1 - 6w_2 + 4w_3' - 4w_4'$$

$$\text{subject to } A'w \leq c'$$

$$\begin{bmatrix} 1 & -2 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 0 & -6 & 3 & -3 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3' \\ w_4' \end{bmatrix} \leq \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix}$$

$$w_1 - 2w_2 + w_3' - w_4' \leq 0$$

$$w_1 - w_2 - w_3' + w_4' \leq 2$$

$$-6w_2 + 3w_3' - 3w_4' \leq 5$$

$$\text{setting } w_3 = w_3' - w_4'$$

$$\text{Maximize } Z' = 2w_1 - 6w_2 + 4w_3$$

$$\text{subject to}$$

$$w_1 - 2w_2 + w_3 \leq 0$$

$$w_1 - w_2 - w_3 \leq 2$$

$$-6w_2 + 3w_3 \leq 5$$

$$w_1, w_2 \geq 0 \text{ and } w_3 \text{ is unrestricted in sign}$$

Solution No 4:

Maximize $Z=10x_1$

subject to

$$x_1 + x_2 \leq 4$$

$$x_1 + x_3 \leq 3$$

$$x_4 \leq 5$$

$$-x_1 \leq 0$$

$$-x_2 \leq 0$$

$$-x_3 \leq 0$$

$$-x_4 \leq 0$$

x_1, x_2, x_3 and x_4 are unrestricted in sign

Solution No 1:

Total demand = $4 + 6 + 8 = 18$

Total Supply = $5 + 4 + 9 = 18$

Since total supply is equal to total demand

Given problem has the initial basic feasible solution.

Solution No 2:

Total demand = $30 + 25 + 35 + 40 = 130$

Total Supply = $50 + 40 + 70 = 160$

Since total supply is not equal to total demand

Given problem has not initial basic feasible solution.

Solution No 3:

Total demand = $5 + 5 + 8 + 4 = 22$

Total Supply = $7 + 7 + 5 = 19$

Since total demand is greater than total supply the given problem is unbalance.

Therefore introduce a dummy row at a unit transportation cost of zero and having the availability of 3 items. The modified table is

	1	2	3	4	Supply
A	x_{11}	x_{12}	x_{13}	x_{14}	7
B	x_{21}	x_{22}	x_{23}	x_{24}	7
C	x_{31}	x_{32}	x_{33}	x_{34}	5
D	0	0	0	0	3
Demand	5	5	8	4	

Now, Total supply = $7 + 7 + 5 + 3 = 22 = 5 + 5 + 8 + 4 =$ Total demand

Given problem is balanced and has the initial basic feasible solution

Solution No 1:

Examine the cost cell carefully and choose the cell with the lowest cost and allocate there as much as possible.

	W₁	W₂	W₃	Supply
F₁	2	7	4	5
F₂	3	3	1(8)	0
F₃	5	4	7	7
F₄	1	6	2	14
Demand	7	9	10	

Since F₂ is exhausted so eliminate that row

	W₁	W₂	W₃	Supply
F₁	2	7	4	5
F₃	5	4	7	7
F₄	1(7)	6	2	7
Demand	0	9	10	

Since W₁ is exhausted so eliminate that column

	W₂	W₃	Supply
F₁	7	4	5
F₃	4	7	7
F₄	6	2(7)	0
Demand	9	3	

Since F₄ is exhausted so eliminate that row

	W₂	W₃	Supply
F₁	7	4	5
F₃	4(7)	7	0
Demand	2	3	

Since F_3 is exhausted so eliminate that row

	W_2	W_3	Supply
F_1	7(2)	4(3)	5
Demand	2	3	

Solution No 2:

	D_1	D_2	D_3	D_4
P_1	6(6)	4(8)	1	5
P_2	8	9(2)	2(14)	7
P_3	4	3	6(1)	2(4)

Total transportation cost by this method is

$$=(6*6)+(4*8)+(9*2)+(2*14)+(6*1)+(2*4)$$

$$=36+32+18+28+6+8=68+46+14=128$$

Solution No 3:

	1	2	3	Supply	Penalties
A	2	7	4	5	(2)
B	3	3	1(8)	8	(2)
C	5	4	7	7	(1)
D	1	6	2	14	(1)
Demand	7	9	18	34	
	(1)	(1)	(1)		

	1	2	3	Supply	Penalties
A	2	7	4	5	(2)
C	5	4	7	7	(1)
D	1	6	2(10)	14	(1)
Demand	7	9	18		
	(1)	(2)	(2)		

	1	2	3	Supply	Penalties
A	2	7	4	5	(2)
C	5	4	7	7	(1)
D	1	6	2(10)	14	(1)
Demand	7	9	10		
	(1)	(2)	(2)		

	1	2	Supply	Penalties
A	2	7	5	(5)
C	5	4	7	(1)
D	1(4)	6	4	(5)
Demand	7	9		
	(1)	(2)		

	1	2	Supply	Penalties
A	2(3)	7(7)	5	(5)
C	5	4(7)	7	(1)
Demand	3	9		
	(3)	(3)		

Thus we get the required feasible solution as shown in the following table

	1	2	3	Supply
A	2(3)	7(2)	4	5
B	3	3	1(8)	8
C	5	4(7)	7	7
D	1(4)	6	2(10)	14
Demand	7	9	18	34

The total transportation cost to this feasible solution

$$=6+14+8+28+4+20=80$$

Solution No 4:

Matrix of cost of allotted cells

			11
15	16		21
	25	16	39

To determine u_i 's and v_j 's we arbitrarily assign one of the multiplier say u_2 equal to zero

			11	u_1
15	16		21	u_2
	25	16	39	u_3
v_1	v_2	v_3	v_4	

$$u_1 + v_4 = 11 \Rightarrow u_1 = 11 - v_4 \Rightarrow u_1 = 11 - 21 = -10 \Rightarrow u_1 = -10$$

$$u_2 + v_1 = 15 \Rightarrow v_1 = 15 - u_2 \Rightarrow v_1 = 15$$

$$u_2 + v_2 = 16 \Rightarrow v_2 = 16 - u_2 \Rightarrow v_2 = 16$$

$$u_2 + v_4 = 21 \Rightarrow v_4 = 21 - u_2 \Rightarrow v_4 = 21$$

$$u_3 + v_2 = 25 \Rightarrow u_3 = 25 - v_2 \Rightarrow u_3 = 25 - 16 = 9 \Rightarrow u_3 = 9$$

$$u_3 + v_3 = 16 \Rightarrow v_3 = 16 - u_3 \Rightarrow v_3 = 16 - 9 = 7 \Rightarrow v_3 = 7$$

Cost matrix of unoccupied cells

19	14	23	
		12	
30			39

u_i and v_j for unoccupied cells

$$u_1 + v_1 = -10 + 15 = 5$$

$$u_1 + v_2 = -10 + 16 = 6$$

$$u_1 + v_3 = -10 + 7 = -3$$

$$u_2 + v_3 = 0 + 7 = 7$$

$$u_3 + v_1 = 9 + 15 = 24$$

$$u_3 + v_4 = 9 + 21 = 30$$

Calculate $c_{ij} - (u_i + v_j)$ for each of unoccupied cells

14	8	26	
		5	
6			9

Since all the values are positive therefore the solution is optimal.

Solution No 5:

Do it by yourself.

Solution No 1:

To determine u_i 's and v_j 's we arbitrarily assign one of the multiplier say u_4 equal to zero

$$u_1 + v_1 = 2 \quad \Rightarrow \quad u_1 = 2 - v_1 \quad \Rightarrow \quad u_1 = 2 - 1 \quad \Rightarrow \quad u_1 = 1$$

$$u_2 + v_3 = 1 \quad \Rightarrow \quad u_2 = 1 - v_3 \quad \Rightarrow \quad u_2 = 1 - 2 = -1$$

$$u_3 + v_2 = 4 \quad \Rightarrow \quad u_3 = 4 - v_2 \quad \Rightarrow \quad u_3 = 4 - 6 = -2$$

$$u_4 + v_1 = 1 \quad \Rightarrow \quad v_1 = 1 - u_4 \quad \Rightarrow \quad v_1 = 1$$

$$u_4 + v_2 = 6 \quad \Rightarrow \quad v_2 = 6 - 0 = 6$$

$$u_4 + v_3 = 2 \quad \Rightarrow \quad v_3 = 2 - u_4 \quad \Rightarrow \quad v_3 = 2 - 0 = 2 \quad \Rightarrow \quad v_3 = 2$$

Since $P_{ij} = C_{ij} - (u_i + v_j)$

$$P_{12} = C_{12} - (u_1 + v_2) = 7 - 7 = 0$$

$$P_{12} = 0$$

$$P_{13} = C_{13} - (u_1 + v_3) = 4 - 3 = 1$$

$$P_{13} = 1$$

Solution No 2:**First Step:**

	Man				
Task	1	2	3	4	5
A	0	2	1	2	4
B	1	3	2	0	3
C	2	3	0	1	2
D	2	0	3	1	0
E	0	4	5	4	2

Second Step:

	Man				
Task	1	2	3	4	5
A	0	2	1	2	4
B	1	3	2	0	3
C	2	3	0	1	2
D	2	0	3	1	0
E	0	4	5	4	2

Solution No 3:**Step 1:**

Subtracting the smallest element of each row from every element of the corresponding row ,we get the following matrix

	Man			
Job	1	2	3	4
A	<u>0</u>	14	9	3
B	9	20	<u>0</u>	22
C	23	<u>0</u>	3	0×
D	9	12	14	<u>0</u>

Step No 2:

Subtracting the smallest element of each column from every element of the corresponding column, we get the following matrix

	Man			
Job	1	2	3	4
A	<u>0</u>	14	9	3
B	9	20	<u>0</u>	22
C	23	<u>0</u>	3	0×
D	9	12	14	<u>0</u>

Step No 3:

Now we test whether it is possible to make an assignment using the zeros by the method described in the handouts

Starting with the row 1 we write in bold and underline(make assignment)in the row containing only one zero and cross the zeros in the corresponding column in which the bold and underline letter lies .Thus we make the following table

	Man			
Job	1	2	3	4
A	<u>0</u>	14	9	3
B	9	20	<u>0</u>	22
C	23	<u>0</u>	3	0×
D	9	12	14	<u>0</u>

Again starting with the column 1 we write in bold and underline (make assignment)in the column containing only one unmarked or uncrossed zero in the above table and cross out the zeros in the corresponding row in which the assignment is marked. Thus we make the following table

	Man			
Job	1	2	3	4
A	<u>0</u>	14	9	3
B	9	20	<u>0</u>	22
C	23	<u>0</u>	3	0×
D	9	12	14	<u>0</u>

Since in the last table every row and every column have one assignment, so we have the complete optimal zero assignment.

Job	A	B	C	D
Man	1	2	3	4

Which is the optimal assignment.

Solution No 4:

Since the row 3 and column 5 have no assignments so we proceed to next step

In this step we draw minimum number of lines to cover all the zeros at least once .For this we proceed as follows:

- 1) We mark (*) row 3 in which there is no assignment.
- 2) Then we mark (*) column 2 and 6 which have zeros in marked row 3
- 3) Then we mark (*) rows 5 and 2 which have assignment in the marked column 2 and 6
- 4) Then we mark column 1 (not already marked) which has zero in the marked row 2.
- 5) Then we mark row 6 which has assignment in the marked column 1.
- 6) Now we draw lines through all marked column 1,2,6. Then we draw lines through unmarked row 1 and 4 having zeros through which there is no line .Thus we, get five lines (minimum number) to cover all the lines.

4	17	49	0	0	17
0	35	25	1	6	0
13	0	59	3	3	0
51	19	0	20	2	4
25	0	42	5	0	2
0	53	46	22	0	20

Now the smallest of the element that do not have a line through them is 4. subtracting this element 4 from all the elements that do not have a line through them and adding to every element that lies at the intersection of two lines and leaving the remaining elements unchanged, we get the following matrix.

4	17	49	0	0	17
0	35	25	1	6	0
13	0	59	3	3	0
51	19	0	20	2	4
25	0	42	5	0	2
0	53	46	22	0	20

Solution No 1:

Years	1	2	3	4	5	6	7
Maintenance cost	600	700	800	900	1000	1200	1500
Cumulative cost	600	1300	2100	3000	4000	5200	6700

Solution No 2:

Cost of Machine = 7000

Capital Cost=Value –Resale value

Years.	1	2	3	4	5	6	7	8
Resale Value	4000	2000	1200	600	500	400	400	400
Capital Cost.	3000	5000	5800	6400	6500	6600	6600	6600

Solution No 3:

Years	1	2	3	4	5	6	7
Maintenance cost	600	700	800	900	1000	1200	1500
Resale value	2000	1333	1000	750	500	300	300
Cumulative cost	600	1300	2100	3000	4000	5200	6700
Capital Cost	1000	1667	2000	2250	2500	2700	2700
Total Cost	1600	2967	4100	5250	6500	7900	9400
Average Cost	1600	1483.5	1366.67	1312.5	1300	1316.67	1342.85